Evidence-Driven Dynamic Visual Selector for Efficient Long Video Understanding

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Abstract. Recent advancements in MLLM-based long-form video understanding have mitigated inference-time computational cost and limited context lengths by selecting query-relevant frames. However, existing approaches predominantly rely on external proxy scorers and rigid heuristic rules, inevitably suffering from misalignment with the target MLLM's intrinsic evidence and failing to accommodate the non-uniform spatiotemporal information density. In this paper, we propose a fine-grained dynamic visual selection framework named **EviSelect**, grounded in the target MLLM's internal attention evidence. Our method efficiently probes visual evidence via self-sparse prefilling as a structured prior to guide distribution-aware dynamic sampling. Specifically, we rapidly approximate attention maps of the target MLLM using highly compressed visual inputs and sparse attention, well-aligned to the full counterpart. Conditioned on three complementary attention components derived from this prior, we design a lightweight selector that not only precisely locates query-relevant timestamps but also adaptively adjusts the local sampling rate and spatial resolution. To enable evidence-conditioned spatiotemporal sampling, we formulate the selector as a stochastic policy and optimize it via GRPO under a joint accuracy–efficiency reward. By rewarding correct predictions under lower visual cost through group-relative comparisons, our method encourages the policy to allocate computation dynamically according to the information density of each video. Extensive experiments across three long video understanding benchmarks demonstrate that EviSelect achieves superior performance compared to existing methods while utilizing only 50% of visual inputs, accelerating the visual selection process by 4.4×.

Keywords: Speculative Prefill · Visual Selection · Long-form Video Understanding

1 Introduction

Long-form video understanding [6, 8, 34] focuses on interpreting events on extended time scales and modeling long-range temporal dependencies in complex video content, enabling critical applications ranging from movie analysis [38] to video surveillance [10]. Although Multimodal Large Language Models (MLLMs) [3, 16] have demonstrated impressive performance in understanding

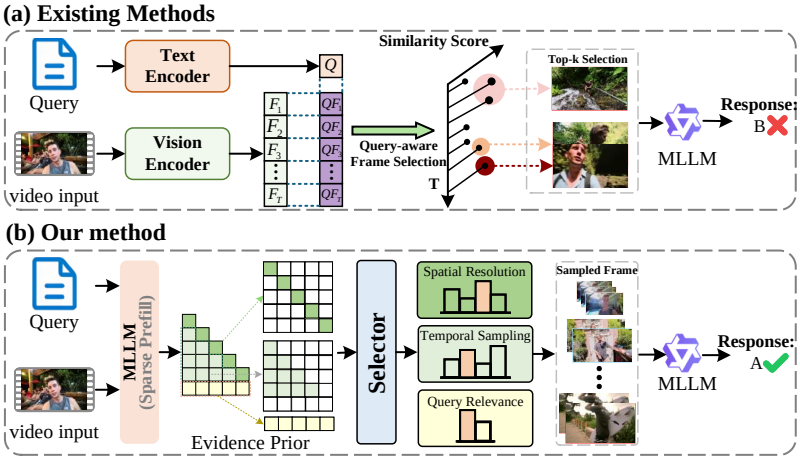

Fig. 1: Comparison of existing method with our method. Existing frame selection method [27, 62] typically rely on CLIP to compute query–frame similarity and apply rigid heuristic rules to select keyframes for MLLM. Our method rapidly approximates the target MLLM's attention maps as evidence priors and designs a selector to predict three policy distributions for each timestamp: retention decision, local sampling rate, and spatial resolution, yielding multi-scale input to generate correct answer.

short video clips, extending their capabilities to long-form videos faces a bottleneck. Due to the limited context windows and attention dilution [35] in long contexts, MLLMs frequently overlook query-relevant evidence and waste computation on redundant frames.

To mitigate this, recent studies [54, 55], as illustrated in Fig. 1(a), typically employ heuristic frame selection strategies driven by external proxies to extract query-conditioned frames. For example, Q-Frame [62] utilizes CLIP [36] to score query-frame similarity, performs ranking-based frame selection, and feeds the selected frames into an MLLM with a fixed resolution strategy. However, these paradigms suffer from three critical limitations: (1) *Objective Misalignment*. Because proxy signals merely reflect semantic relevance rather than aligning with the target MLLM's internal evidence requirements [23], existing paradigms [15,24] inevitably miss subtle but critical visual cues. (2) *Static Frame Retrieval*. Existing methods [27, 47] often treat long-video understanding as an isolated keyframe selection problem, rather than modeling a query-conditioned belief distribution over continuous temporal trajectories. Consequently, these approaches lead to temporally myopic selections that over-concentration on visually salient segments while insufficient coverage of sparse yet critical events. (3) *Rigid Spatiotemporal Configuration*. Existing methods [43, 56] adopt a one-size-fits-all sampling setup—selecting a predetermined number of frames at a fixed spatial resolution—regardless of the underlying non-uniform information density of the video. Such rigid configurations result in inefficient allocation of the com-

putational budget, either over-sampling redundant segments or under-sampling sparse critical events.

To alleviate these challenges, we propose **EviSelect**, a fine-grained dynamic visual selection framework, as shown in Fig. 1 (b). Rather than relying on external proxies, EviSelect directly extracts endogenous evidence from attention maps, which inherently reflect its query-conditioned spatiotemporal dependencies and actual visual demands. Specifically, we first employ a sparse pre-filling mechanism over highly compressed video inputs to efficiently probe the target MLLM's internal attention maps. By isolating query-to-vision, cross-frame, and intra-frame attention weights, we decouple this prior into three specialized structural cues that explicitly capture semantic relevance, temporal dependencies, and spatial saliency. A lightweight selector then leverages them to drive a joint distribution-aware sampling policy by predicting three corresponding fine-grained dimensions at each timestamp: retention decision, local sampling rate, and spatial resolution. To align this sampling policy with the target MLLM's actual performance, we formulate the selector as a stochastic policy and optimize it via GRPO [40] using a joint accuracy-efficiency reward. By rewarding correct MLLM predictions achieved at lower visual cost, our method drives the policy to adaptively allocate computation according to the video's intrinsic spatiotemporal information density, rather than relying on fixed heuristics.

Our main contributions can be summarized as follows:

- We propose EviSelect, a fine-grained dynamic visual selection framework that reformulates keyframe selection as a distribution-aware dynamic sampling driven by internal evidence of the target MLLM.
- We propose a sparse prefill mechanism that rapidly approximate the authentic evidence prior and a selector for fine-grained, distribution-aware spatiotemporal sampling policy.
- Extensive experiments on three benchmarks demonstrate that our method achieves state-of-the-art performance with only 50% of the visual inputs and accelerates the visual selection process by 4.4×.

2 Related Work

2.1 Long-form Video Understanding.

Existing MLLMs [2, 9, 16] have made significant progress in vision-language understanding, with representative early works tracing back to BLIP-2 [20] and Flamingo [1]. However, understanding long-form videos remains challenging for the MLLM due to its limited context window and prohibitive computational cost. To address this limitation, recent studies typically follow two paradigms: (1) token compression [5, 45], motivated by the idea that there are large amounts of visual redundancy between frames, introduces trainable compression modules and fine-tunes the model to aggregate visual content into fewer tokens [22, 44]. For example, Videochat-flash [21] propose a hierarchical token compression strategy that compress long video from clip-level to video-level. (2) context extension [42], which enables MLLMs to accept longer inputs via length extrapolation

and further training [7, 60] on longer video sequences. Long-VITA [7] implement context-parallelism distributed inference and logits-masked language modeling head to scale up the infinite number of tokens. Despite their effectiveness, they typically rely on additional model modifications, and often struggle to preserve key information in the video.

2.2 Query-based Keyframe Selection.

Instead of modifying the MLLM via compression or context extension, a complementary research direction for long-form video understanding focuses on selecting query-relevant keyframes to capture critical visual information, ranging from training-free frame matching [62, 67] to training-based selectors [4, 65]. Training-free keyframe selection methods [27, 47] typically rely on external proxies, such as CLIP [36] or DINOv2 [33], to select the informative frames. For example, DIG [19] adapts its sampling strategy based on the query type and uses DINOv2 [33] to compute pairwise frame similarity to derive candidate segments, and then leverages a MLLM-based reward signal to expand local segments. AKS [47] leverages CLIP [36] to compute frame–query relevance scores and selects keyframes by jointly optimizing relevance and temporal coverage. Meanwhile, training-based selectors [15, 56] model frame selection as a learnable policy. Frame-Voyager [58] constructs subset-level supervision to query frame subsets combinationally rather than selecting frames independently. TSPO [46] proposes a trainable event-aware temporal agent, which captures event-query correlation for probabilistic keyframe selection. FFS [4] proposes a flexible and controllable frame selection operation by introducing a novel register token to the selector. Despite their effectiveness, they rely on external proxy scorers and rigid heuristic rules, which suffers from misalignment with the intrinsic preferences of the target MLLM, and often fails to accommodate the dynamic spatiotemporal information density.

2.3 Reinforcement Learning for MLLMs.

Recent reinforcement learning methods [17, 61] have advanced beyond classic RLHF pipelines [37, 39] by adopting group-based policy optimization objectives for more stable MLLM optimization. In particular, Group Relative Policy Optimization (GRPO) [40] estimates advantages through within-group relative ranking among multiple sampled candidates for the same input within each update, reducing reliance on critic training and improving robustness under high-variance rewards. Such group-based RL formulations have been explored in a range of applications, especially for tasks emphasizing strong reasoning [51] or structured decision-making [11], where relative preference signals are easier to obtain and more stable than absolute scalar rewards. In this paper, our EviSelect takes a novel perspective that directly leverages GRPO to optimize dynamic sampling strategy that balances accuracy and token efficiency, which directly addresses the most critical bottleneck in long video understanding: extracting decisive evidence from extremely long contexts.

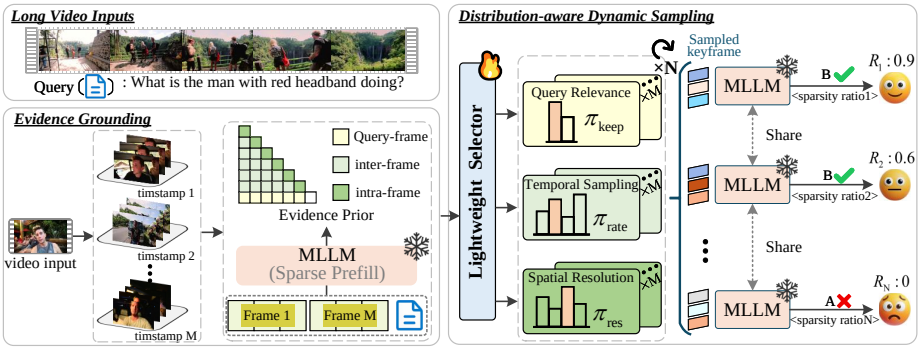

Fig. 2: Overview of our method. Our method efficiently probes visual evidence to enable distribution-aware dynamic sampling. We partition the video into uniform timestamps, construct a highly compressed visual input, and employ sparse attention to rapidly approximate the MLLM's attention maps. A lightweight selector then predicts three policy distributions for each timestamp, including retention decision, sampling rate, and spatial resolution. These decisions are used to resample the video for final inference. The policy is optimized via GRPO with a joint accuracy–efficiency reward.

3 Method

3.1 Overview

Given a textual query Q and a long video V, we aim to construct a compact yet informative multi-scale visual input V_s that aligns with the target MLLM's intrinsic evidence demands and adapts to the video's dynamic information density. We cast this process as a distribution-aware spatiotemporal sampling policy that outputs fine-grained visual allocation decisions $\mathbf{u}_t = (z_t, \rho_t, \lambda_t)$ at each timestamp t, representing timestamp retention, local temporal sampling rate, and spatial resolution, respectively. As illustrated in Fig. 2, our framework efficiently captures the target MLLM's attention priors and models the joint sampling policy. In the evidence grounding stage, we perform an efficient probing pass through the self-sparse prefilling of the target MLLM over a highly compressed version of V, obtaining attention maps $\mathbf{A}$ as structured priors. Then we decompose $\mathbf{A}$ into three complementary cues: query-frame relevance $\mathbf{A}_{qf}$ for z_t, frame-frame temporal dependency $\mathbf{A}_{ff}$ for ρ_t, and intra-frame spatial saliency $\mathbf{A}_{if}$ for λ_t. Conditioned on these cues, a lightweight selector directly predicts the joint distribution over $\mathbf{u}_t$, and is trained through GRPO with a joint accuracy-efficiency reward. As a result, the trained selector dynamically adjusts the fine-grained decisions, extracting only the essential visual context. Finally, we resample V according to these decisions to produce V_s for final inference by the target MLLM.

3.2 Evidence Grounding

We extract layer-wise structural attention priors from the target MLLM via an efficient probing pass on a highly compressed video. Specifically, we perform self-

sparse prefilling to obtain block-level sparse attention maps $\{A^{(l)}\}_{l=1}^{L}$ for all L layers, which are then aggregated at the frame level and organized into three cues $\mathbf{A}_{qf}$, $\mathbf{A}_{ff}$, and $\mathbf{A}_{if}$.

Sparse Evidence Probing. Since the sampling decisions (z_t, ρ_t, λ_t) operate at the frame level, it suffices to approximate the attention distribution of the target MLLM at a relatively coarse granularity, circumventing the heavy computational cost of reconstructing dense token-level attention over the full video. Therefore, we first obtain a highly compressed yet representative frame subset V_c from the original video V through joint temporal and spatial downsampling, and then concatenate the given text query Q and V_c to form the prefilling context sequence X. During the prefilling stage, we employ a block-sparse attention mechanism [31] coupled with a top-p retention strategy [13, 26], effectively providing a reliable and efficient estimate of the attention distribution, with minimal deviation from dense attention. Formally, this efficient probing process can be abstracted as:

$$\{A^{(l)}\} = \text{SparsePrefill}(X; p, B), \tag{1}$$

where p denotes the cumulative probability threshold for retention, B indicates the block size used by block-sparse attention.

Evidence Decomposition. We aggregate block-level sparse attention $A^{(l)}$ into frame-level and stack the resulting maps across all L layers to form the layer-wise evidence $\mathbf{A}$. We then partition $\mathbf{A}$ into three complementary components, as shown in Fig. 2(middle), to explicitly guide the dynamic sampling policy. Let $T = |V_c|$ denote the number of timestamps (anchor frames) in V_c: $\mathbf{A}_{qf} \in \mathbb{R}^{L\times T}$ models query-frame relevance by aggregating attention from query tokens to the visual tokens of each anchor frame; $\mathbf{A}_{ff} \in \mathbb{R}^{L\times T\times T}$ captures the temporal dependency across timestamps by aggregating attention between visual tokens across anchor-frames pairs; $\mathbf{A}_{if} \in \mathbb{R}^{L\times T}$ measures intra-frame spatial complexity via attention concentration within each anchor frame.

Due to causal masking, $\mathbf{A}_{ff}$ is strictly lower-triangular; as a result, different amounts of valid past context varies across timestamps, which can introduce systematic position-dependent bias in the aggregated temporal attention [53]. We mitigate this bias following [49], yielding a position-robust prior. The resulting set of structural cues $\{\mathbf{A}_{qf}, \mathbf{A}_{ff}, \mathbf{A}_{if}\}$ provides comprehensive guidance for the subsequent selector.

3.3 Distribution-Aware Dynamic Sampling

Given the evidence priors that are extracted from the target MLLM, we model visual selection as a distribution-aware sampling policy over the timestamp sequence. Concretely, the selector takes $\mathbf{A}_{qf}$, $\mathbf{A}_{ff}$, and $\mathbf{A}_{if}$ as inputs, and predicts three timestamp-wise categorical policy distributions: π_{query} over timestamp retention $z_t \in \{0, 1\}$, π_{rate} over local sampling rate $\rho_t \in \mathcal{R}$, and π_{res} over spatial

resolution $\lambda_t \in \mathcal{L}$, where $\mathcal{R}$ and $\mathcal{L}$ denote predefined candidate sets. Specifically, we first aggregate $\mathbf{A}_{ff}$ to reduce its shape $L \times T$. Then we all these $L \times T$ attention into a unified feature space using separate linear projections, fuse the projected features, and feed them into a shared lightweight MLP to obtain hidden representations $\mathbf{h}$. Then three independent prediction heads output the categorical distributions $\pi_{\text{query}}(\cdot \mid \mathbf{h})$, $\pi_{\text{rate}}(\cdot \mid \mathbf{h})$, and $\pi_{\text{res}}(\cdot \mid \mathbf{h})$; each head produces a length-T set of timestamp-wise distributions in one forward pass.

The predicted decisions are applied on the original video V at each selected timestamp: for $z_t = 1$, we resample frames within the temporal segment represented by timestamp t according to ρ_t and resize the sampled frames based on λ_t, yielding the final multi-scale frame sequence V_s for downstream MLLM inference. The joint probability of sampling policy is formulated as:

$$\pi_s(V_s \mid Q, V_c) = \prod_{t=1}^{T} \pi_{\text{query}}(z_t \mid \mathbf{h})\ \pi_{\text{rate}}(\rho_t \mid \mathbf{h})\ \pi_{\text{res}}(\lambda_t \mid \mathbf{h}), \tag{2}$$

3.4 Policy Optimization with GRPO

The reward of our method is composed of performance and efficiency rewards. Specifically, the performance reward function R_{per} is binary: a correct answer is assigned to 1 and otherwise 0 [46]. We define the efficiency reward R_{eff} as a weighted combination of a retention term and a token-cost term:

$$R_{\text{eff}} = \eta \left(1 - \frac{\sum_{t=1}^{T} z_t}{T}\right) + (1-\eta)\left(1 - \frac{C_{V_s}}{C_{\max}}\right), \tag{3}$$

where η balances the retention and budget terms; C_{V_s} is the number of visual tokens in V_s; and $C_{\max}$ is a per-candidate upper bound computed on the same retained timestamps by applying the maximum sampling rate and resolution.

However, a naive additive combination of performance and efficiency can become efficiency-biased when no candidate in the group is correct, because the accuracy signal collapses to a constant while the efficiency term still pushes budgets downward. To probe the efficiency–accuracy trade-off without collapsing to a trivial low-cost policy, we apply the efficiency reward only when the sampling group contains at least one correct answer. Concretely, given a sampling group $\{\mathbf{o}_1, \ldots, \mathbf{o}_G\}$, we define a group-level indicator:

$$C \triangleq \mathbb{1}\{\exists j \in \{1, \ldots, G\} : R_{\text{per}}(\mathbf{o}_j) = 1\}. \tag{4}$$

The overall per-sample reward for the i-th candidate is then defined as:

$$R_i = R_{\text{per},i} + \alpha \cdot C \cdot R_{\text{eff},i}, \tag{5}$$

where α controls the strength of the efficiency reward. Finally, we optimize the selector policy using GRPO by maximizing the following objective:

$$J(\theta) = \mathbb{E}_{Q,V_c}\left[\frac{1}{G}\sum_{i=1}^{G} \min\Big(r_i(\theta)\,A_i,\ \text{clip}(r_i(\theta), 1 \pm \epsilon)\,A_i\Big) - \beta\, D_{\text{KL}}\Big(\pi_{s,\theta} \,\big\|\, \pi_{\text{ref}}\Big)\right], \tag{6}$$

where $A_i = \frac{R_i - \text{mean}(R_1, R_2, \ldots, R_G)}{\text{std}(R_1, R_2, \ldots, R_G)}$ is the group-relative advantage computed from the group rewards, and $r_i(\theta) = \frac{\pi_{s,\theta}(V_s^{(i)}|Q, V_c)}{\pi_{s,\theta_{\text{old}}}(V_s^{(i)}|Q, V_c)}$ denotes the importance ratio defined as the probability of sampling $V_s^{(i)}$ under the current selector policy divided by that under the old policy. $D_{\text{KL}}\Big(\pi_{s,\theta} \,\Big\|\, \pi_{\text{ref}}\Big)$ measures the KL divergence between the updated policy and reference policies, and β is the weight coefficient of the KL divergence term.

4 Experiments

4.1 Benchmarks

To evaluate the effectiveness of our method, we conduct extensive experiments on three widely used long-form video understanding benchmarks: (1) MLVU [66], a multi-task benchmark that ranges from 3 minutes to 2 hours (average video duration: 12 minutes) and covers diverse video genres such as movies and surveillance footage. We evaluate on the "M-Avg" portion of the "Dev" split, following [63]. (2) LongVideoBench [52], evaluated on the validation set without subtitles (average video duration: 12 minutes), containing 1,337 QA pairs across 17 fine-grained categories. (3) Video-MME [12] comprises 900 videos with durations split into short (< 2min), medium (4∼15min), and long (30∼60min), totaling 2,700 QA pairs (average video duration: 17 minutes).

4.2 Implementation Detail

We evaluate our framework on two MLLMs: Qwen2.5-VL-7B [3] and LLaVA-Video-7B [64]. All experiments are conducted on 4 NVIDIA H100 GPUs. Following TSPO [46], our lightweight selector (0.1M parameters) is trained on the same 10K subset of LLaVA-Video-178K dataset [64] for one epoch, with a learning rate of 5×10^{-4} and a batch size of 1. In the GRPO algorithm, we set the efficiency reward coefficient $\alpha = 0.1$, the retention–budget trade-off $\eta = 0.5$, and the KL coefficient $\beta = 0.04$. We set the number of tokens per frame to $N_v = 20$ during evidence grounding. We uniformly sample timestamps every 2 seconds. The top-p threshold is set to $\tau_p = 0.97$ and the block size is fixed at $B = 20$. To align with the intrinsic temporal patch size of the visual encoder (e.g., a temporal patch size of 2 in Qwen2.5-VL [3]), we define the candidate set for sampling rate as $\mathcal{R} = \{2, 4, 8, 16\}$. The spatial resolution candidate set $\mathcal{L}$ comprises 4 discrete levels ranging from 90×160 to 1080×1920.

4.3 Main Results

Comparison with existing methods. As shown in Tab. 1, LLaVA-Video-7B equipped with EviSelect achieves state-of-the-art performance across three general long video benchmarks, while using substantially fewer visual tokens. Compared with TSPO [46], our EviSelect achieves performance improvements of 1.9%

Table 1: Evaluation of long-form video understanding on three commonly used benchmarks. The three benchmarks are evaluated using lmms-eval [59]. "*" denotes our reproduced results under 16 frames.

Model	Frames	LLM Size	Selector	LongVideoBench	MLVU	Video-MME	
				Val	Dev	Long	Avg
Close-Source MLLM							
GPT-4o [16]	-	-	Uniform	66.7	64.6	65.3	71.9
GPT-4V [32]	-	-	Uniform	61.3	49.2	53.5	59.9
Gemini-1.5-Flash [48]	-	-	Uniform	61.6	-	61.1	70.3
Gemini-1.5-Pro [48]	-	-	Uniform	64	-	67.4	75.0
Open-Source MLLM							
Video-LLaVA [25]	8	7B	Uniform	-	36.2	-	39.9
Qwen-VL [2]	8	7B	Uniform	-	-	37.8	41.1
Oryx-1.5 [30]	64	7B	Uniform	56.3	-	51.2	58.8
LLaVA-Onevision [18]	32	7B	Uniform	56.4	64.7	46.7	58.2
NVILA [29]	1024	7B	Uniform	57.7	70.1	54.8	64.2
Apollo [68]	2FPS	7B	Uniform	58.5	68.7	-	61.3
mPLUG-Owl3 [57]	128	7B	Uniform	59.7	70.0	50.1	59.3
LongVU [41]	1FPS	7B	DINOv2	-	65.4	-	60.6
Qwen2-VL [50]	8	7B	Uniform	53.5	56.9	45.3	53.7
LLaVA-Video-7B*	16	7B	Uniform	57.4	64.4	51.3	60.3
+AKS [47]	16	7B	CLIP	61.6	-	-	62.2
+TSPO [46]*	16	7B	TSPO-0.4B	61.4	68.2	52.9	62.1
+EviSelect	**8.1**	7B	**0.1B**	**62.1**	**69.1**	**53.7**	**64.0**
Qwen2.5-VL-7B*	16	7B	Uniform	55.4	54.3	48.9	57.1
+Q-Frame [62]	16	7B	CLIP	57.37	56.81	49.02	-
+DIG [19]	16	7B	DINOv2	57.89	63.98	51.93	-
+TSPO [46]*	16	7B	TSPO-0.4B	58.1	65.1	52.1	59.1
+EviSelect	**7.9**	7B	**0.1B**	**59.2**	**66.1**	**53.0**	**60.8**

on Video-MME [12], 0.9% on MLVU [66], and 0.7% on LongVideoBench [52], while using only 8.1 frames on average, reducing the frame budget by 49% compared with TSPO (16 frames). Moreover, EviSelect using Qwen2.5-VL-7B [3] also brings consistent improvements over TSPO, demonstrating strong generalization across different MLLM architectures. Meanwhile, it also achieves competitive performance in the challenging "Long" subset of Video-MME. This indicates its robust ability to localize query-relevant content even under extremely long contexts, which benefits from our use of the evidence priors of the target MLLM to dynamically identify key timestamps.

Inference Efficiency. We fix the number of candidate frames to 720 to eliminate the effect of varying video lengths for a fair efficiency comparison. As demonstrated in Tab. 2, compared with TSPO [46], our method reduces the keyframe selection time from 9.2s to 2.1s. In addition, by selecting fewer yet

Table 2: Inference efficiency comparison with existing frame selection methods.

Method	Frames	Token	Frame Selection Time	MLLM Inference Time
LLaVA-Video	720→16	3360	0	0.78
LLaVA-Video+CoS [14]	128→64	13440	28.4	2.7
LLaVA-Video+TSPO [46]	720→16	3360	9.2	0.78
LLaVA-Video+EviSelect	720→8.1	**1701**	**2.1**	**0.45**

more informative frames, we preserve the accuracy gains over the baseline while requiring nearly half the visual tokens (1701 vs. 3360) and reducing the MLLM inference time to 58% of TSPO. Notably, Tab. 3 shows that while the average video duration expands by over 30× from the "Short" to "Long" subsets in Video-MME, the visual token consumption of our method scales by only 3.1× (from 0.9k to 2.8k). This sub-linear growth demonstrates that our method effectively localizes query-relevant visual content and adaptively filters out massive visual redundancy, rather than blindly accumulating uninformative frames.

Table 3: Token cost across video lengths.

Method	Video-MME		
	Short	Medium	Long
Time	1.3min	9min	41min
Token	0.9k	1.8k	2.8k

4.4 Ablation Study

Effect of Different Components. To evaluate the contribution of each component in EviSelect, we conduct comprehensive ablation studies on MLVU [66] and LongVideoBench [52] using LLaVA-Video-7B [64] as the MLLM. As shown in Tab. 4, removing the retention strategy and reverting to uniform sampling incurs the largest efficiency loss and degrades accuracy, dropping from 62.3% to 60.3% on MLVU and from 58.5% to 57.4% on LongVideoBench. This indicates that evidence-driven retention retention of key timestamps is crucial for discarding redundant segments while preserving critical evidence, whereas uniform sampling frequently misses query-relevant visual cues.

Moreover, disabling adaptive temporal sampling or enforcing a uniform original resolution results in significant accuracy degradations of 3.7% and 1.2% on MLVU [66], respectively. This performance gap occurs because, without adaptive sampling and resolution allocation, this variant fails to accommodate the varying information distribution of temporal and spatial complexity. Notably, adaptive resolution offers a better accuracy–cost trade-off: it achieves higher accuracy with 2048 tokens than adaptive sampling with 2730 tokens. This indicates that temporal continuity of critical evidence is vital for long-form video understanding, while the semantic information of most frames remains remarkably preserved even at low resolutions. Combining all components achieves the best performance with the lowest cost.

Table 4: Ablation study of different components on 2 benchmarks.

Retention Strategy	Adaptive Sampling	Adaptive Resolution	Token	LongVideoBench	MLVU
✗	✗	✗	3360	57.4	60.3
✓	✗	✗	2100	58.5	62.3
✓	✓	✗	2730	59.1	65.4
✓	✗	✓	2048	60.3	67.9
✓	✓	✓	**1701**	**62.1**	**69.1**

Table 5: Evaluation on TempCompass [28] for temporal perception.

Method	Frame	TempCompass					
		Action	Direction	Speed	Event Order	Attribute Change	Avg.
TSPO	16	92.6	59.4	64.9	63.7	72.3	72.1
TSPO	32	93.1	61.3	66.7	65.4	74.0	73.9
Ours	15.6	**94.2**	**62.9**	**69.1**	**66.5**	**75.6**	**75.4**

Timestamp Sampling Interval. Fig. 4 demonstrates that a 2-second timestamp interval is sufficient for reliable evidence grounding to improve performance, and using 1s brings only a marginal improvement of 0.2% but doubles the token cost. This suggests that the MLLM can infer the missing temporal dynamics between sampled timestamps and approximate a continuous spatiotemporal evidence distribution from moderately discrete observations. In contrast, coarser intervals lead to a significant accuracy drop, as sparse timestamps provide insufficient temporal anchors for capturing short-lived events and abrupt state changes.

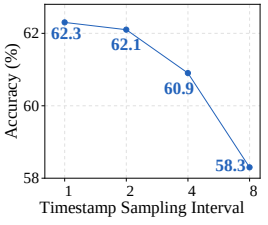

Fig. 4: Ablation study of timestamp interval on LongVideoBench [52].

Adaptability to Temporal Dynamics. To evaluate the dynamic temporal perception ability of EviSelect, we further conduct experiments on TempCompass [28], whose videos exhibit stronger temporal dynamics and denser temporal information. In Tab. 5, EviSelect achieves 75.4% accuracy with only 15.6 sampled frames, outperforming TSPO [46] using 32 frames by 1.5%. The improvement is consistent across all sub-tasks, with particularly notable gains on Speed and Attribute Change, indicating that EviSelect can better capture fine-grained variations by adaptively increase sampling density in rapidly changing segments.

Number of Per-frame Token N_v. The number of tokens per frame directly determines the computational cost of the evidence grounding. As shown in the

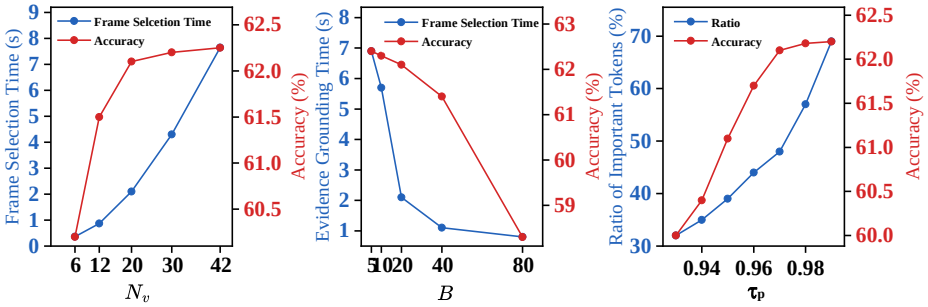

Fig. 3: Influence of per-frame token number N_v, block size B and top-p threshold τ_p on LongVideoBench [52].

left of Fig. 3, we study its impact on LongVideoBench [52]. We can observe that increasing N_v improves accuracy at the cost of longer frame selection time, but the gains quickly saturate while overly aggressive compression leads to substantial performance drops. Based on this trend, $N_v = 20$ offers a good balance between accuracy and efficiency. Moreover, these results reveal a key principle: evidence grounding focuses on frame-level temporal change rather than fully reconstructing high-fidelity visual details. And the compressed visual sequence typically can provide reliable evidence prior.

Effect of Block Size B. To further investigate the impact of the block granularity, we vary the block size $B \in \{5, 10, 20, 40, 80\}$, given that each frame is represented by 20 visual tokens. The results are illustrated in the middle of Fig. 3. Intuitively, smaller B yields better performance, suggesting that smaller block size enables more finer-grained evidence prior, whereas larger blocks tend to mix tokens across different frames and blur the temporal boundaries of evidence. Nevertheless, we adopt $B = 20$ as the default setting, since reducing B further incurs a significant increase in frame selection time but brings only marginal accuracy gains. This is because our goal is to preserve evidence priors at the frame level and enforce overly token-level granularityoften retains unnecessary fine detail, which incurs additional evidence grounding overhead without improving evidence quality.

Effect of τ_p in Top-p Attention. As demonstrated in Fig. 3, we analyze how varying the token retention threshold τ_p affects both the model performance and the ratio of important tokens on LongVideoBench [52]. In general, a lower retention threshold leads to a reduced ratio for important tokens, thereby enhancing the inference efficiency at the cost of performance degradation, since the preserved context is insufficient to maintain a reliable evidence prior. Conversely, the gains from increasing τ_p quickly saturate. Based on this trend, the optimal range for τ_p lies around 0.97, which offers a good balance between accuracy and efficiency.

Table 6: Ablation study of different sampling rate on LongVideoBench [52].

Sampling Rate				Token	Acc
High	Medium	Low	Tiny		
✓				3045	60.7
	✓			2048	60.3
		✓		1624	58.5
			✓	1356	55.2
✓	✓			2458	61.0
✓	✓	✓		1950	61.5
✓	✓	✓	✓	**1701**	**62.1**

Table 7: Ablation study of different resolution on LongVideoBench [52].

Spatial Resolution				Token	Acc
High	Medium	Low	Tiny		
✓				4261	59.3
	✓			2735	59.1
		✓		2059	57.5
			✓	1563	54.9
✓	✓			3218	60.2
✓	✓	✓		2879	61.1
✓	✓	✓	✓	**1701**	**62.1**

Multi-rate Sampling Strategy. Tab. 6 highlights the effect of various sampling rate strategies on LongVideoBench [52]. The results show that using all four sampling rate levels achieves the highest accuracy with a lower token cost. Compared with the best single-rate configuration, the four-level setting improves accuracy of 1.4% while reducing token cost from 3045 to 1701. Moreover, as the number of available sampling rate levels increases, the accuracy consistently improves, suggesting that a fixed sampling rate cannot simultaneously handle fast actions and slowly changing scenes effectively and multi-rate sampling can adapt more flexibly to the varying temporal complexity across different video segments.

Multi-resolution Strategy. As shown in Tab. 7, We further investigate the impact of different resolution combination strategies on LongVideoBench [52]. We observe that enabling all resolution levels not only achieves the highest accuracy of 62.1% but also maintains a lower token cost, indicating that effective resolution design is not uniformly increasing image quality, but adaptively allocating resolution on demand. In contrast, using only tiny resolution frames significantly reduces the token cost but lowers the accuracy to 54.9%. Relying solely on high-resolution frames also fails to achieve optimal performance, with accuracy reaching only 59.3% and the token cost rising to the highest value of 4261. These results suggest that low-resolution settings are efficient but tend to lose critical subtle visual evidence, whereas high-resolution settings preserve more details but also enlarge large amounts of background, introducing extra token and visual distraction.

4.5 Qualitative Results.

Fig. 5 illustrates the visualization results of keyframe selection and model responses. In the left example, when recognizing tiny text such as a battery model number, uniform sampling method is blind to the query, selecting frames from various irrelevant scenes. In contrast, our method not only precisely localizes the key time segments but also adaptively assigns a higher resolution to the selected

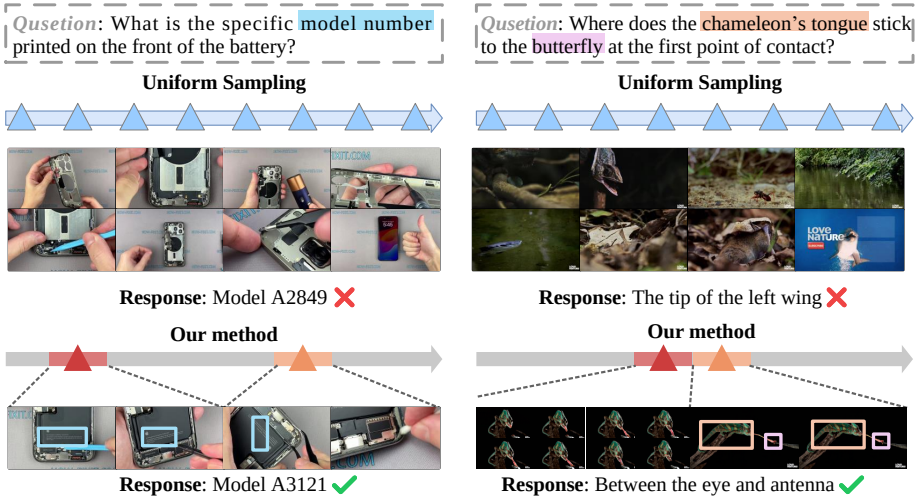

Fig. 5: Qualitative comparison between uniform sampling and our method.

frames to preserve fine-grained details, achieving the correct answer with fewer frames. In the right example, the query relies on the critical moment when the chameleon shoots its tongue and catches the butterfly. Our method adaptively densifies sampling around this action while using a lower resolution to control cost, thereby capturing continuous and complete visual evidence at a lower overhead. These examples confirm that our method leverages the MLLM's internal evidence priors to effectively localize query-relevant timestamps, adaptively increases resolution to recognize subtle details, and adjusts the sampling rate to capture rapidly changing targets.

5 Conclusion

In this paper, we present **EviSelect**, a framework that leverage target MLLM's internal attention priors for distribution-aware spatiotemporal sampling. By self-sparse prefilling on compressed inputs, EviSelect efficiently approximates attention evidence, based on which a lightweight selector predicts retention, local sampling rate, and spatial resolution policies. Optimized end-to-end via GRPO under a joint accuracy–efficiency reward, this probabilistic policy dynamically aligns visual inputs with the target MLLM's intrinsic requirements. Experiments on three long-video benchmarks show that EviSelect consistently outperforms prior methods while using only 50% visual inputs and achieving up to 4.4× acceleration in visual selection.

Limitations. Since evidence priors are extracted from the internal attention of the target MLLM, our EviSelect may have limited generalization when transferred to other MLLMs. Moreover, it is not applicable to closed-source MLLMs where internal attentions are inaccessible.

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